# OIP v1.0.27 Addendum: Mathematical Provenance, Operationalization Gaps & Completion Register

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## 1 Overview

This document establishes the mathematical provenance for OIP v1.0.27. Its primary function is to register specification-defined mathematics, initial calibration parameters, operational assumptions, and unresolved operational definitions without introducing unauthorized numerical substitutes.

## 2 Specification-Defined Mathematical Structures

The following mathematical structures are explicitly present in the OIP v1.0.26 specification. Their presence in the specification does not, by itself, constitute empirical validation, independent derivation, or external verification:

- $II = 1 - IDS$
- OIP Stability $= GFL \times II \times AML$
- $LIV = \dfrac{RI \times SR \times AR}{IDS}$
- Score $= 0.6 \times OE + 0.3 \times CV + 0.1 \times VIS$
- $SIS = 0.5 \times RC + 0.3 \times AFL + 0.2 \times CAS$
- $IS = 0.1 \times SD + 0.9 \times CW$

Under the current LIV expression, numerical evaluation is undefined at $IDS = 0$.

## 3 Initial Calibration Parameters

The following numerical coefficients are treated as initial calibration parameters. Empirical calibration and statistical derivation remain pending:

- Score weights: $(0.6, 0.3, 0.1)$
- SIS weights: $(0.5, 0.3, 0.2)$
- Stress-model weights: $(0.1, 0.9)$

## 4 Operational Assumptions and Pilot Parameters

The following values are operational choices rather than mathematically derived constants:

- $IDS$ threshold: $IDS > 0.7$
- LIN pilot size: 5 participants
- Pilot duration: 30 days

## 5 Operationally Unspecified Quantities

The following quantities require explicit measurement protocols and functional definitions before reproducible operational measurement:

- $GFL = f(\text{RealWorld\_Outcome\_Error\_Correction\_Rate})$
- Exact operational definition of $AML$ (Adaptive Moral Logic)
- Measurement protocol for $IDS$
- Measurement protocols for $OE$, $CV$, $VIS$
- Measurement protocols for $RC$, $AFL$, $CAS$
- Measurement protocols for $RI$, $SR$, $AR$

## 6 Boundary Conditions

For the LIV expression,

$$LIV = \frac{RI \times SR \times AR}{IDS} \tag{1}$$

at $IDS = 0$, the current LIV formulation is mathematically **undefined**.

No numerical substitute, epsilon floor ($\epsilon$), or replacement rule is introduced in v1.0.27. A boundary-handling rule remains separately unspecified.

## 7 Validation Status

The mathematical structures documented in this register are specification-level models. Their presence in the specification does not constitute empirical, economic, production, or hardware validation.

## 8 Completion Status

At the v1.0.27 addendum stage, the documented mathematical structures are registered at the specification level. Calibration, measurement operationalization, empirical validation, and independent reproduction remain open work items.

## 9 Validation Pathway

The proposed pathway for validation follows a structured progression:

Operationalization → Reference Implementation → Unit Tests
→ Synthetic Benchmarks → E1 Experiments → Independent Reproduction